

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Is the [resultant] force for circular motion (1) It is directed towards the centre [of the circle] (1)	2			2		
	(b)	(i)		$T = \frac{15}{10} = 1.5 \text{ s}$ or $f = \frac{10}{15} = 0.67 \text{ [Hz]}$ (1) $v = \frac{2\pi r}{T} = \frac{2\pi(0.8)}{1.5} (1)[= 3.35 \text{ m s}^{-1}]$ Alternative: Total distance = $1.6\pi \times 10$ (1) $v = \frac{1.6\pi \times 10}{15} (1) [= 3.35 \text{ m s}^{-1}]$	1	1		2	2	
		(ii)		Substitution: $F = \frac{mv^2}{r} = \frac{(30 \times 10^{-3}) \times 3.35^2}{0.8} (1)$ $= 0.42 \text{ [N]} (1)$ Alternative: $F = mr\omega^2 = (30 \times 10^{-3}) \times 0.8 \times \left(\frac{2\pi}{1.5}\right)^2 (1)$ $= 0.42 \text{ [N]}(1)$	1	1		2	2	
	(c)			Force = Tension + mg (1) Tension = Force – $mg = 0.42$ ecf – $(30 \times 10^{-3})(9.81) = 0.13 \text{ [N]}$ (1) Award 1 mark for 0.7 [N]		2		2	1	

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	(d)	(i)		Conservation of energy or implied (1) Full substitution or good algebra e.g. $\frac{1}{2}mv_b^2 = mgh + \frac{1}{2}mv^2$ OR Initial KE = 0.17 [J] and PE loss = 0.47 [J] (implied by final KE = 0.64 J) OR $v_b = \sqrt{2gh + v^2}$ (1) Final velocity = 6.53 [m s⁻¹] (1) Alternative: Use of $v^2 = u^2 + 2ax$ with $u = 3.35$ m s⁻¹ and $a = 9.81$ m s⁻² (1) Correct answer = 6.53 [m s⁻¹] (1) Some statement of why this equation works e.g. due to conservation of energy, we can consider a particle dropping with initial downward speed of 3.35 (1) Accept – this equation shouldn't work but it does! Alternative: Use of $v^2 = u^2 + 2ax$ with $u = 0$ gives $v = 5.6$ [m s⁻¹] (1) Pythagoras applied i.e. $\sqrt{3.35^2 + 5.6^2}$ (1) Correct answer = 6.53 [m s⁻¹] (1)		3		3	2	
		(ii)		Use of $x = ut + \frac{1}{2}at^2$ for the vertical motion or 2 other usable equations (1) $t = \sqrt{\frac{2x}{g}} = \sqrt{\frac{2(1.2+0.8)}{9.81}} = 0.64$ [s] (1) For the horizontal motion: horizontal distance from point of released = $vt = 3.35 \times 0.64$ ecf (1) = 2.14 [m] so claim is correct (1) ecf Accept $2.14 \approx 2$			4	4	3	
				Question 2 total	4	7	4	15	10	0